

Grade 7 lessons

Signed numbers, proportions and two-step equations

IN THIS BOOK

1. Multiplying signed numbers
2. Proportions
3. Percent change
4. Two-step equations
5. Circles and chance

HOW TO USE IT

Read one lesson together, then practice it. Each lesson matches 5 questions in the Grade 7 Kahoot and Blooket sets, and one stop on the Grade 7 trail in the Math Trail app.

Every lesson has the big idea, the steps, two worked examples, one mistake to watch for, and the words to know.

Multiplying signed numbers

THE BIG IDEA

When you multiply or divide two numbers, the signs decide whether the answer is positive or negative.

HOW TO DO IT

1. Ignore the signs and multiply or divide the numbers.
2. If the two signs are the same, the answer is positive.
3. If the two signs are different, the answer is negative.
4. With more steps, follow the order of operations: multiply and divide before you add and subtract.

WORKED EXAMPLES

$$(-3) \times (-4) = ?$$

$3 \times 4 = 12$. The signs are the same, so the answer is positive: 12.

$$(-20) \div 5 = ?$$

$20 \div 5 = 4$. The signs are different, so the answer is negative: -4.

WATCH OUT

The sign rules for multiplying are not the rules for adding. $-3 + (-4)$ is -7 , but $-3 \times (-4)$ is $+12$.

WORDS TO KNOW

product

the answer to a multiplication

quotient

the answer to a division

Proportions

THE BIG IDEA

A proportion says two ratios are equal. If you know three of the four numbers, you can find the fourth.

HOW TO DO IT

1. Write the two ratios as fractions with the same kind of thing on top in both.
2. Look for the number one side was multiplied by, and do the same to the other part.
3. Or cross-multiply: multiply each top by the opposite bottom, then solve.
4. In $y = kx$, k is the constant: divide y by x to find it.

WORKED EXAMPLES

Solve $\frac{3}{4} = \frac{x}{20}$.

4 was multiplied by 5 to get 20. So $x = 3 \times 5 = 15$.

5 notebooks cost \$8. How much do 15 cost?

15 is 3 times as many as 5. So the cost is $3 \times 8 = \$24$.

WATCH OUT

Keep the same things in the same places. If notebooks are on top in one ratio, they must be on top in the other.

WORDS TO KNOW

proportion

two equal ratios

constant of

the number k in $y = kx$

proportionality

Percent change

THE BIG IDEA

Discounts, tax and tips are all percents of a price. Percent change compares how much something changed with where it started.

HOW TO DO IT

1. Find the percent of the price by multiplying.
2. For a discount, subtract that amount from the price.
3. For tax or a tip, add that amount to the price.
4. Percent change = $\text{change} \div \text{original amount} \times 100$.

WORKED EXAMPLES

A \$40 jacket is 25% off. What is the sale price?

25% of 40 is 10. $40 - 10 = \$30$.

A price goes from \$40 to \$50. What is the percent increase?

The change is 10. $10 \div 40 = 0.25$, which is 25%.

WATCH OUT

Divide by the original amount, not the new one. From 40 to 50 is a 25% increase, not 20%.

WORDS TO KNOW

discount

money taken off a price

percent change

the change compared with the starting amount

Two-step equations

THE BIG IDEA

A two-step equation has two things done to x . Undo them in reverse order to get x on its own.

HOW TO DO IT

1. Undo the adding or subtracting first.
2. Then undo the multiplying or dividing.
3. Do the same thing to both sides every time.
4. If there are brackets, you can divide both sides by the number in front first.
5. Check by putting your answer back in.

WORKED EXAMPLES

Solve $3x + 5 = 20$.

Subtract 5 from both sides: $3x = 15$.

Divide both sides by 3: $x = 5$. Check: $3 \times 5 + 5 = 20$.

Solve $2(x + 3) = 14$.

Divide both sides by 2: $x + 3 = 7$.

Subtract 3: $x = 4$.

WATCH OUT

Whatever you do to one side, do to the whole of the other side. Do not divide only one term.

WORDS TO KNOW

inverse operation

the operation that undoes another

Circles and chance

THE BIG IDEA

Every circle is described by its radius. Probability measures how likely something is, from 0 (never) to 1 (certain).

HOW TO DO IT

1. The radius is the distance from the center to the edge. The diameter is twice the radius.
2. Circumference (distance around) = $2\pi r$.
3. Area = πr^2 .
4. Probability = number of ways you want $\div$ number of all possible ways.

WORKED EXAMPLES

A circle has radius 5. Find its circumference and area.

Circumference: $2 \times \pi \times 5 = 10\pi$. Area:
 $\pi \times 5 \times 5 = 25\pi$.

A bag has 3 red and 5 blue marbles. What is the chance of picking red?

There are 8 marbles and 3 are red.
The chance is $\frac{3}{8}$.

WATCH OUT

Squaring is not doubling. 5^2 is 25, not 10. And if you are given the diameter, halve it first.

WORDS TO KNOW

radius

center to edge

circumference

the distance around a circle

probability

how likely something is